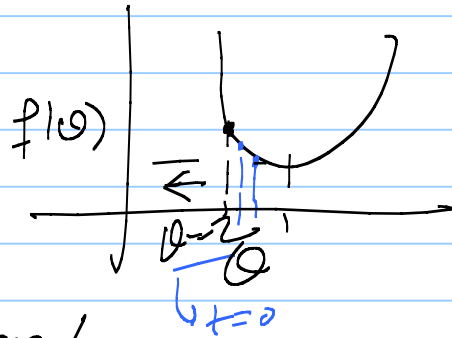


Machine Learning

Aug 8, 2025

Last class:-



Gradient descent:-

$$\nabla f(\theta) \big|_{\theta=2} =$$

$$2(\theta - 3) \big|_2 = -2$$

$$f(\theta) = (\theta - 3)^2 + 1$$

$$\min_{\theta} f(\theta)$$

At local minima/maxima:-

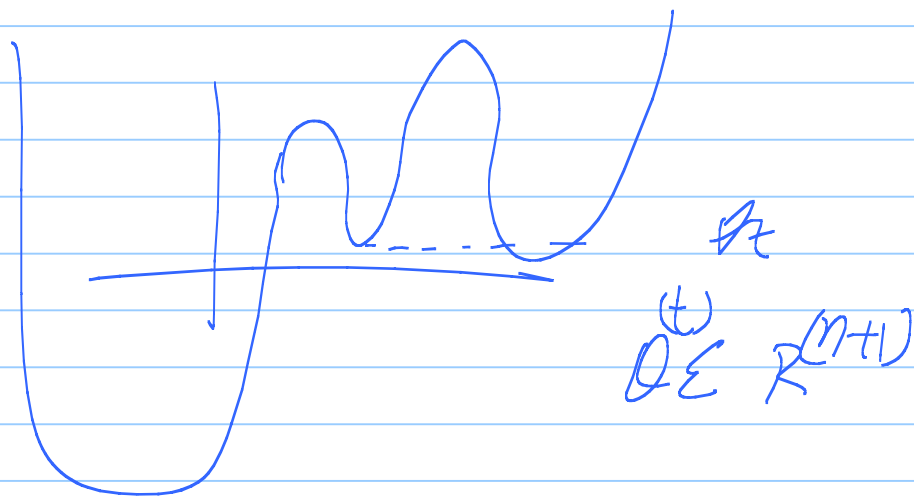
$$\frac{\partial f(\theta)}{\partial \theta} \bigg|_{\theta^*} = 0$$

$$\hookrightarrow \theta \text{ s.t. } \nabla f(\theta) = 0$$

$$\begin{array}{l} t=0 \quad \theta^{(1)} \leftarrow 2 \quad \eta = 0.1 \\ \theta^{(1)} \leftarrow 2 - 0.1(-2) = 2.2 \quad \theta^{(2)} \leftarrow 2.2 - 0.1(-0.8) = 2.38 \end{array}$$

$\theta^{(0)}$ $\uparrow$ init Iterative Algorithm - $\nabla_{\theta} f(\theta) |_{\theta^{(t)}}$
 $\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \left(\frac{\partial f(\theta)}{\partial \theta} \right) |_{\theta^{(t)}}$
 η $\nwarrow$ Learning Rate

$$\nabla_{\theta} f(\theta) = \begin{bmatrix} \frac{\partial f(\theta)}{\partial \theta_0} \\ 1 \\ \frac{\partial f(\theta)}{\partial \theta_1} \end{bmatrix} \quad \theta \in \mathbb{R}^2 \quad \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 0$$



Gradient Descent
 1. $t \leftarrow 0$;
 2. $\theta^{(t)} \leftarrow$ initializes $\text{init}()$;

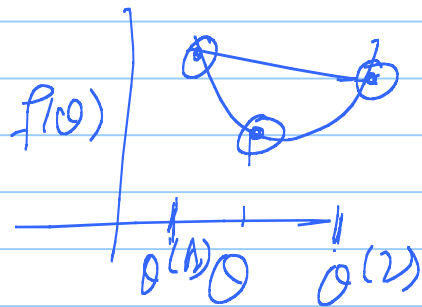
do

$$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \nabla_{\theta} f(\theta) |_{\theta^{(t)}}$$

$t \leftarrow t+1$;

3 while ! converged)

↳ Convex Functions :-



⇒ f is convexly differentiable.

Hessian

$$\Rightarrow \begin{cases} \frac{\partial^2 f(\theta)}{\partial \theta^2} \geq 0 \end{cases}$$

⇒ if f is convexly differentiable
 $\theta^{(1)}, \theta^{(2)} \in \mathbb{R}$

convex

$$0 \leq \alpha \leq 1 \Leftrightarrow f(\underbrace{\alpha \theta^{(1)} + (1-\alpha) \theta^{(2)}}_{\text{convex combination}}) \leq \alpha f(\theta^{(1)}) + (1-\alpha) f(\theta^{(2)})$$